

Vocabulary Types for Composite Class Design

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Abstract

We propose the addition of two new class-templates to the C++ Standard Library: `indirect<T>` and `polymorphic<T>`.

These class-templates have value semantics and compose well with other standard library types (such as `vector`) allowing the compiler to correctly generate special member functions.

The class template, `indirect`, confers value-like semantics on a free-store-allocated object. An `indirect` may hold an object of a class `T`, copying the `indirect` will copy the object `T`. When a parent object contains a member of type `indirect<T>` and is accessed through a `const` access path, `constness` will propagate from the parent object to the instance of `T` owned by the `indirect` member.

The class template, `polymorphic`, confers value-like semantics on a free-store allocated object. A `polymorphic<T>` may hold an object of a class publicly derived from `T`, copying the `polymorphic<T>` will copy the object of the derived type. When a parent object contains a member of type `polymorphic<T>` and is accessed through a `const` access path, `constness` will propagate from the parent object to the instance of `T` owned by the `polymorphic` member.

This proposal is a fusion of two older individual proposals, P1950 and P0201. The design of the two class-templates is sufficiently similar that they should not be considered in isolation.

Motivation

The standard library has no vocabulary type for a free-store-allocated object with value semantics. When designing a composite class we may need an object to be stored indirectly to support incomplete types, reduce object size or support open-set polymorphism.

We propose two new additions to the standard library to represent indirectly stored values: `indirect` and `polymorphic`; they represent free-store allocated

objects with value-semantics. `polymorphic<T>` can own any object of a type publicly derived from `T` allowing composite classes to contain polymorphic components. We require the addition of two classes to avoid the cost of virtual dispatch (calling the copy constructor of a potentially derived-type object through type-erasure) when copying of polymorphic objects is not needed.

Design requirements

We review the fundamental design requirements of `indirect` and `polymorphic` that make them suitable for composite class design.

Special member functions

Both class templates should be suitable for use as members of composite classes where the compiler will generate special member functions. This means that the class templates should provide the special member functions where they are supported by the owned object type `T`.

- `indirect<T, Alloc>` and `polymorphic<T, Alloc>` are default constructible in cases where `T` is default constructible.
- `indirect<T, Alloc>` is copy constructible where `T` is copy constructible and assignable.
- `polymorphic<T, Alloc>` is unconditionally copy constructible and assignable.
- `indirect<T, Alloc>` and `polymorphic<T, Alloc>` are unconditionally move constructible and assignable.
- `indirect<T, Alloc>` and `polymorphic<T, Alloc>` destroy the owned object in their destructors.

Deep copies

Copies of `indirect<T>` and `polymorphic<T>` should own copies of the owned object created with the copy constructor of the owned object. In the case of `polymorphic<T>` this means that the copy should own a copy of a potentially derived type object created with the copy constructor of the derived type object.

Note: Including a `polymorphic` component in a composite class means that virtual dispatch will be used (through type-erasure) in copying the `polymorphic` member. Where a composite class contains a `polymorphic` member from a known set of types, prefer `std::variant` or `indirect<std::variant>` if indirect storage is required.

const propagation

When composite objects contain `pointer`, `unique_ptr` or `shared_ptr` members they allow non-const access to their respective pointees when accessed through a

const access path. This prevents the compiler from eliminating a source of const-correctness bugs and makes it difficult to reason about the const-correctness of a composite object.

Accessors of unique and shared pointers do not have const and non-const overloads:

```
T* unique_ptr<T>::operator->() const;
T& unique_ptr<T>::operator*() const;
```

```
T* shared_ptr<T>::operator->() const;
T& shared_ptr<T>::operator*() const;
```

When a parent object contains a member of type `indirect<T>` or `polymorphic<T>`, access to the owned object (of type T) through a const access path should be `const` qualified.

```
struct A {
    enum class Constness { CONST, NON_CONST };
    Constness foo() { return Constness::NON_CONST; }
    Constness foo() const { return Constness::CONST; };
};

class Composite {
    indirect<A> a_;

    Constness foo() { return a_.foo(); }
    Constness foo() const { return a_.foo(); };
};

int main() {
    Composite c;
    assert(c.foo() == A::Constness::NON_CONST);
    const Composite& cc = c;
    assert(cc.foo() == A::Constness::CONST);
}
```

Value semantics

Both `indirect` and `polymorphic` are value-types whose owned object is free-store-allocated (or some other memory-resource controlled by the specified allocator).

When a value type is copied it gives rise to two independent objects that can be modified separately.

The owned object is part of the logical state of `indirect` and `polymorphic`. Operations on a const-qualified object do not make changes to the object's logical state nor to the logical state of other object.

`indirect<T>` and `polymorphic<T>` are default constructible in cases where `T` is default constructible. Moving a value type onto the free-store should not add or remove the ability to be default constructed.

Pairwise-comparison operators, which are defined only for `indirect`, compare the owned objects where the owned objects can be compared: where `T` is ordered, `indirect<T>` is also ordered.

The hash operation, which is defined only for `indirect`, hashes the owned object where the owned object can be hashed.

We discuss why only `indirect` is comparable and hashable in an appendix.

Unobservable null state and interaction with `std::optional`

Both `indirect` and `polymorphic` have a null state which is used to implement move. The null state is not intended to be observable to the user. There is no `operator bool` or `has_value` member function. Accessing the value of an `indirect` or `polymorphic` after it has been moved from is erroneous behaviour. We provide a `valueless_after_move` member function that returns `true` if an object is in a valueless state to allow explicit checks for the valueless state in cases where it cannot be verified statically.

Without a null state, moving `indirect` or `polymorphic` would require allocation and moving from the owned object. This would be expensive and would require the owned object to be moveable. The existence of a null state allows move to be implemented cheaply without requiring the owned object to be moveable.

Where a nullable `indirect` or `polymorphic` is required, using `std::optional` is recommended. This may be commonplace as `indirect` and `polymorphic` may be used in composite classes where smart pointers are currently used to (mis)represent component objects. Putting `T` onto the free-store should not make it nullable. Nullability must be explicitly opted-into by using `std::optional<indirect<T>>` or `std::optional<polymorphic<T>>`.

`std::optional<>` is specialized for `indirect<>` and `polymorphic<>` so they incur no additional overhead.

Access to a `std::optional<indirect<T>>` or `std::optional<polymorphic<T>>` can be done with double indirection, `(**v)`, or with a single arrow operator to access a member, `v->some_member`.

Note: As the null state of `indirect` and `polymorphic` is not observable, and access to a moved-from object is erroneous, `std::optional` can be specialized by implementers to exchange pointers on move construction and assignment.

Design for polymorphic types

To be used as a base class with `polymorphic`, a type `PolymorphicInterface` does not need a virtual destructor. The same mechanism that is used to call

the copy constructor of a potentially derived-type object will be used to call the destructor.

To allow compiler generation of special member functions of an abstract interface type `PolymorphicInterface` in conjunction with `polymorphic`, `PolymorphicInterface` needs at least a non-virtual protected destructor and a protected copy constructor. `PolymorphicInterface` does not need to be assignable, move constructible or move assignable for `polymorphic<PolymorphicInterface>` to be assignable, move constructible or move assignable.

```
class PolymorphicInterface {
protected:
    PolymorphicInterface(const PolymorphicInterface&) = default;
    ~PolymorphicInterface() = default;
public:
    // virtual functions
};
```

For an interface type with a public virtual destructor, users would potentially pay the cost of virtual dispatch twice when deleting `polymorphic<I>` objects containing derived-type objects.

All derived types owned by a `polymorphic` must be publicly copy-constructible.

Prior work

This proposal continues the work started in [P0201] and [P1950].

Previous work on a cloned pointer type [N3339] met with opposition because of the mixing of value and pointer semantics. We feel that the unambiguous value semantics of `indirect` and `polymorphic` as described in this proposal address these concerns.

Impact on the standard

This proposal is a pure library extension. It requires additions to be made to the standard library header `<memory>`.

Technical specifications

X.Y Class template `indirect` [indirect]

X.Y.1 Class template `indirect` general [indirect.general] An *indirect value* is an object that manages the lifetime of an owned object. An indirect value object is *valueless* if it has no owned object. An indirect value may only become valueless after it has been moved from. Every object of type `indirect<T, Allocator>` uses an object of type `Allocator` to allocate and free storage for the owned `T` object as needed.

The template parameter `T` of `indirect` must be a non-union class type.

The template parameter `T` of `indirect` may be an incomplete type.

X.Y.2 Class template `indirect` synopsis [indirect.syn]

```
template <class T, class Allocator = std::allocator<T>>
class indirect {
    T* p_; // exposition only
    Allocator allocator_; // exposition only
public:
    using value_type = T;
    using allocator_type = Allocator;

    constexpr indirect();

    template <class... Ts>
    explicit constexpr indirect(std::in_place_t, Ts&&... ts);

    template <class... Ts>
    constexpr indirect(
        std::allocator_arg_t, const Allocator& alloc, std::in_place_t, Ts&&... ts);

    constexpr indirect(const indirect& other);

    constexpr indirect(
        std::allocator_arg_t, const Allocator& alloc, const indirect& other);

    constexpr indirect(indirect&& other) noexcept;

    constexpr indirect(
        std::allocator_arg_t, const Allocator& alloc, indirect&& other) noexcept;

    constexpr ~indirect();

    constexpr indirect& operator=(const indirect& other);

    constexpr indirect& operator=(indirect&& other) noexcept(see below);

    constexpr const T& operator*() const noexcept;

    constexpr T& operator*() noexcept;

    constexpr const T* operator->() const noexcept;

    constexpr T* operator->() noexcept;
```

```

constexpr bool valueless_after_move() const noexcept;

constexpr allocator_type get_allocator() const noexcept;

constexpr void swap(indirect& other) noexcept(see below);

friend constexpr void swap(indirect& lhs, indirect& rhs) noexcept(see below);

template <class U, class AA>
friend constexpr bool operator==(
    const indirect<T, A>& lhs, const indirect<U, AA>& rhs);

template <class U, class AA>
friend constexpr bool operator!=(
    const indirect<T, A>& lhs, const indirect<U, AA>& rhs);

template <class U, class AA>
friend constexpr auto operator<=>(
    const indirect<T, A>& lhs, const indirect<U, AA>& rhs);

template <class U>
friend constexpr bool operator==(const indirect<T, A>& lhs, const U& rhs);

template <class U>
friend constexpr bool operator==(const U& lhs, const indirect<T, A>& rhs);

template <class U>
friend constexpr bool operator!=(const indirect<T, A>& lhs, const U& rhs);

template <class U>
friend constexpr bool operator!=(const U& lhs, const indirect<T, A>& rhs);

template <class U>
friend constexpr auto operator<=>(const indirect<T, A>& lhs, const U& rhs);

template <class U>
friend constexpr auto operator<=>(const U& lhs, const indirect<T, A>& rhs);
};

template <class T, class Alloc>
struct std::uses_allocator<indirect<T, Alloc>, Alloc> : true_type {};

template <class T, class Alloc>
struct hash<indirect<T, Alloc>>;

```

X.Y.3 Constructors [indirect.ctor]

`indirect()`

- *Constraints:* `is_default_constructible_v<T>` is true.
- *Effects:* Constructs an indirect owning a default constructed T.
- *Postconditions:* `*this` is not valueless.

`template <class... Ts>`

`explicit constexpr indirect(std::in_place_t, Ts&&... ts);`

- *Constraints:* `is_constructible_v<T, Ts...>` is true.
- *Effects:* Constructs an indirect owning an instance of T created with the arguments Ts.
- *Postconditions:* `*this` is not valueless.

`template <class... Ts>`

`constexpr indirect(`

`std::allocator_arg_t, const Allocator& alloc, std::in_place_t, Ts&&... ts);`

- *Constraints:* `is_constructible_v<T, Ts...>` is true.
- *Preconditions:* `Allocator` meets the *Cpp17Allocator* requirements.
- *Effects:* Equivalent to the preceding constructor except that the allocator is initialized with `alloc`.
- *Postconditions:* `*this` is not valueless.

`constexpr indirect(const indirect& other);`

- *Constraints:* `is_copy_constructible_v<T>` is true.
- *Preconditions:* `other` is not valueless.
- *Effects:* Constructs an indirect owning an instance of T created with the copy constructor of the object owned by `other`.
- *Postconditions:* `*this` is not valueless.

`constexpr indirect(`

`std::allocator_arg_t, const Allocator& alloc, const indirect& other);`

- *Constraints:* `is_copy_constructible_v<T>` is true and `uses_allocator<T, Allocator>` is true.
- *Preconditions:* `other` is not valueless and `Allocator` meets the *Cpp17Allocator* requirements.
- *Effects:* Equivalent to the preceding constructor except that the allocator is initialized with `alloc`.
- *Postconditions:* `*this` is not valueless.


```
constexpr indirect(indirect&& other) noexcept;
```

- *Preconditions:* `other` is not valueless.
- *Effects:* Constructs an `indirect` owning the object owned by `other`.
- *Postconditions:* `other` is valueless.
- *Remarks:* This constructor does not require that `is_move_constructible_v<T>` is true.

```
constexpr indirect(
    std::allocator_arg_t, const Allocator& alloc, indirect&& other) noexcept;
```

- *Constraints:* `is_copy_constructible_v<T>` is true and `uses_allocator<T, Allocator>` is true.
- *Preconditions:* `other` is not valueless and `Allocator` meets the *Cpp17Allocator* requirements.
- *Effects:* Equivalent to the preceding constructors except that the allocator is initialized with `alloc`.
- *Postconditions:* `other` is valueless.
- *Remarks:* This constructor does not require that `is_move_constructible_v<T>` is true.

X.Y.4 Destructor [`indirect.dtor`]

```
constexpr ~indirect();
```

- *Effects:* If `*this` is not valueless, destroys the owned object.

X.Y.5 Assignment [`indirect.assign`]

```
constexpr indirect& operator=(const indirect& other);
```

- *Preconditions:* `other` is not valueless.
- *Effects:* If `*this` is not valueless and `std::is_copy_assignable_v<T>` is true, then the owned object in `*this` is copy assigned from the owned object in `other`. Otherwise if `*this` is not valueless and `std::is_copy_assignable_v<T>` is false, destroys the owned object and then, constructs a new owned object using the copy constructor of the object owned by `other`. Otherwise if `*this` is valueless, constructs an owned object using the copy constructor of the object owned by `other`.
- *Postconditions:* `*this` is not valueless.

```
constexpr indirect& operator=(indirect&& other) noexcept(
    allocator_traits<Allocator>::propagate_on_container_move_assignment::value ||
    allocator_traits<Allocator>::is_always_equal::value);
```

- *Preconditions:* `other` is not valueless.
- *Effects:* If `*this` is not valueless, destroys the owned object. Then takes ownership of the object owned by `other`.
- *Postconditions:* `*this` is not valueless. `other` is valueless.

X.Y.6 Observers [indirect.observers]

```
constexpr const T& operator*() const noexcept;
constexpr T& operator*() noexcept;
```

- *Preconditions:* `*this` is not valueless.
- *Effects:* Returns a reference to the owned object.
- *Remarks:* These functions are constexpr functions.

```
constexpr const T* operator->() const noexcept;
constexpr T* operator->() noexcept;
```

- *Preconditions:* `*this` is not valueless.
- *Effects:* Returns a pointer to the owned object.
- *Remarks:* These functions are constexpr functions.

```
constexpr bool valueless_after_move() const noexcept;
```

- *Returns:* true if `*this` is valueless, otherwise false.

```
constexpr allocator_type get_allocator() const noexcept;
```

- *Returns:* A copy of the Allocator object used to construct the owned object or, if that allocator has been replaced, a copy of the most recent replacement.

X.Y.7 Swap [indirect.swap]

```
constexpr void swap(indirect& other) noexcept(
    allocator_traits<Allocator>::propagate_on_container_swap::value ||
    allocator_traits<Allocator>::is_always_equal::value);
```

- *Preconditions:* `*this` is not valueless, `other` is not valueless.
- *Effects:* Swaps the objects owned by `*this` and `other`.
- *Remarks:* Does not call `swap` on the owned objects directly.

```
constexpr void swap(indirect& lhs, indirect& rhs) noexcept(
    allocator_traits<Allocator>::propagate_on_container_swap::value ||
    allocator_traits<Allocator>::is_always_equal::value);
```

- *Preconditions:* `lhs` is not valueless, `rhs` is not valueless.
- *Effects:* Swaps the objects owned by `lhs` and `rhs`.

- *Remarks:* Does not call `swap` on the owned objects directly.

X.Y.8 Relational operators [indirect.rel]

```
template <class U, class AA>
constexpr bool operator==(const indirect<T, A>& lhs, const indirect<U, AA>& rhs);
```

- *Preconditions:* lhs is not valueless, rhs is not valueless.
- *Effects:* returns `*lhs == *rhs`.
- *Remarks:* Specializations of this function template for which `*lhs == *rhs` is a core constant expression are constexpr functions.

```
template <class U, class AA>
constexpr bool operator!=(const indirect<T, A>& lhs, const indirect<U, AA>& rhs);
```

- *Preconditions:* lhs is not valueless, rhs is not valueless.
- *Effects:* returns `*lhs != *rhs`.
- *Remarks:* Specializations of this function template for which `*lhs != *rhs` is a core constant expression, are constexpr functions.

```
template <class U, class AA>
constexpr auto operator<=>(const indirect<T, A>& lhs, const indirect<U, AA>& rhs);
```

- *Preconditions:* lhs is not valueless, rhs is not valueless.
- *Effects:* returns `*lhs <=> *rhs`.
- *Remarks:* Specializations of this function template for which `*lhs <=> *rhs` is a core constant expression, are constexpr functions.

X.Y.9 Comparison with T [indirect.comp.with.t]

```
template <class T, class A, class U>
constexpr bool operator==(const indirect<T, A>& lhs, const U& rhs);
```

- *Preconditions:* lhs is not valueless.
- *Effects:* returns `*lhs == rhs`.
- *Remarks:* Specializations of this function template for which `*lhs == *rhs` is a core constant expression, are constexpr functions.

```
template <class T, class A, class U>
constexpr bool operator==(const U& lhs, const indirect<T, A>& rhs);
```

- *Preconditions:* rhs is not valueless.
- *Effects:* returns `lhs == *rhs`.
- *Remarks:* Specializations of this function template for which `*lhs == *rhs` is a core constant expression, are constexpr functions.

```
template <class T, class A, class U>
constexpr bool operator!=(const indirect<T, A>& lhs, const U& rhs)
```

- *Preconditions:* lhs is not valueless.
- *Effects:* returns `*lhs != rhs`.
- *Remarks:* Specializations of this function template for which `*lhs != *rhs` is a core constant expression, are constexpr functions.

```
template <class T, class A, class U>
constexpr bool operator!=(const U& lhs, const indirect<T, A>& rhs);
```

- *Preconditions:* rhs is not valueless.
- *Effects:* returns `lhs != *rhs`.
- *Remarks:* Specializations of this function template for which `*lhs != *rhs` is a core constant expression, are constexpr functions.

```
template <class T, class A, class U>
constexpr auto operator<=(const indirect<T, A>& lhs, const U& rhs);
```

- *Preconditions:* lhs is not valueless.
- *Effects:* returns `*lhs <= rhs`.
- *Remarks:* Specializations of this function template for which `*lhs <= *rhs` is a core constant expression, are constexpr functions.

```
template <class T, class A, class U>
constexpr auto operator<=(const U& lhs, const indirect<T, A>& rhs);
```

- *Preconditions:* rhs is not valueless.
- *Effects:* returns `lhs <= *rhs`.
- *Remarks:* Specializations of this function template for which `*lhs <= *rhs` is a core constant expression, are constexpr functions.

X.Y.10 Allocator related traits [indirect.allocator.traits]

```
template <class T, class Alloc>
struct std::uses_allocator<indirect<T>, Alloc> : true_type {};
```

- *Preconditions:* Alloc meets the *Cpp17Allocator* requirements.

X.Y.11 Hash support [indirect.hash]

```
template <class T, class Alloc>
struct std::hash<indirect<T, Alloc>>;
```

- *Preconditions:* i is not valueless.

The specialization `hash<indirect<T, Alloc>>` is enabled ([unord.hash]) if and only if `hash<remove_const_t<T>>` is enabled. When enabled, for an object `i` of type `indirect<T, Alloc>`, then `hash<indirect<T, Alloc>>()(i)` evaluates to the same value as `hash<remove_const_t<T>>()( *i)`. The member functions are not guaranteed to be `noexcept`.

X.Y.12 Optional support [indirect.optional]

```
template <class T, class Alloc>
class std::optional<indirect<T, Alloc>>;
```

The specialization `std::optional<indirect<T, Alloc>>` guarantees `size(std::optional<indirect<T, Alloc>>) == size(indirect<T, Alloc>>)`.

```
// [optional.observe], observers
constexpr const indirect<T, Alloc>& operator->() const noexcept;
constexpr indirect<T, Alloc>& operator->() noexcept;
```

- *Preconditions:* `*this` contains a value.
- *Returns:* `val`.
- *Remarks:* These functions are `constexpr`. The specialization `std::optional<indirect<T, Alloc>>` provides `operator->` that returns a reference to the contained `indirect`.

Otherwise, the interface of the specialization is as defined in [optional].

X.Z Class template polymorphic [polymorphic]

X.Z.1 Class template polymorphic general [polymorphic.general] A *polymorphic value* is an object that manages the lifetime of an owned object. A polymorphic value object may own objects of different types at different points in its lifetime. A polymorphic value object is *valueless* if it has no owned object. A polymorphic value may only become valueless after it has been moved from. Every object of type `polymorphic<T, Allocator>` uses an object of type `Allocator` to allocate and free storage for the owned object as needed.

The template parameter `T` of `polymorphic` must be a non-union class type.

The template parameter `T` of `polymorphic` may be an incomplete type.

X.Z.2 Class template polymorphic synopsis [polymorphic.syn]

```
template <class T, class Allocator = std::allocator<T>>
class polymorphic {
    control_block* control_block_; // exposition only
    Allocator allocator_; // exposition only
public:
    using value_type = T;
    using allocator_type = Allocator;
```

```

polymorphic();

template <class U, class... Ts>
explicit constexpr polymorphic(std::in_place_type_t<U>, Ts&&... ts);

template <class U, class... Ts>
constexpr polymorphic(
    std::allocator_arg_t, const Allocator& alloc, std::in_place_type_t<U>, Ts&&... ts);

constexpr polymorphic(const polymorphic& other);

constexpr polymorphic(
    std::allocator_arg_t, const Allocator& alloc, const polymorphic& other);

constexpr polymorphic(polymorphic&& other) noexcept;

constexpr polymorphic(
    std::allocator_arg_t, const Allocator& alloc, polymorphic&& other) noexcept;

constexpr ~polymorphic();

constexpr polymorphic& operator=(const polymorphic& other);

constexpr polymorphic& operator=(polymorphic&& other) noexcept(see below);

constexpr const T& operator*() const noexcept;

constexpr T& operator*() noexcept;

constexpr const T* operator->() const noexcept;

constexpr T* operator->() noexcept;

constexpr bool valueless_after_move() const noexcept;

constexpr allocator_type get_allocator() const noexcept;

constexpr void swap(polymorphic& other) noexcept(see below);

friend constexpr void swap(polymorphic& lhs, polymorphic& rhs) noexcept(see below);
};

template <class T, class Alloc>
struct std::uses_allocator<polymorphic<T, Alloc>, Alloc> : true_type {};

```

X.Z.3 Constructors [polymorphic.ctor]

`polymorphic()`

- *Constraints:* `is_default_constructible_v<T>` is true, `is_copy_constructible_v<T>` is true.
- *Effects:* Constructs a polymorphic owning a default constructed T.
- *Postconditions:* `*this` is not valueless.

`template <class U, class... Ts>`

`explicit polymorphic(std::in_place_type_t<U>, Ts&&... ts);`

- *Constraints:* `is_base_of_v<T, U>` is true, `is_constructible_v<U, Ts...>` is true, `is_copy_constructible_v<U>` is true.
- *Effects:* Constructs a polymorphic owning an instance of U created with the arguments Ts.
- *Postconditions:* `*this` is not valueless.

`template <class U, class... Ts>`

`polymorphic(`

`std::allocator_arg_t, const Allocator& alloc, std::in_place_type_t<U>, Ts&&... ts);`

- *Constraints:* `is_base_of_v<T, U>` is true, `is_constructible_v<U, Ts...>` is true, `is_copy_constructible_v<U>` is true.
- *Preconditions:* `Allocator` meets the *Cpp17Allocator* requirements.
- *Effects:* Equivalent to the preceding constructor except that the allocator is initialized with `alloc`.
- *Postconditions:* `*this` is not valueless.

`polymorphic(const polymorphic& other);`

- *Preconditions:* `other` is not valueless.
- *Effects:* Constructs a polymorphic owning an instance of T created with the copy constructor of the object owned by `other`.
- *Postconditions:* `*this` is not valueless.

`polymorphic(`

`std::allocator_arg_t, const Allocator& alloc, const polymorphic& other);`

- *Preconditions:* `other` is not valueless and `Allocator` meets the *Cpp17Allocator* requirements.
- *Effects:* Equivalent to the preceding constructor except that the allocator is initialized with `alloc`.
- *Postconditions:* `*this` is not valueless.

`polymorphic(polymorphic&& other) noexcept;`

- *Preconditions*: `other` is not valueless.
- *Effects*: Constructs a polymorphic that takes ownership of the object owned by `other`.
- *Postconditions*: `other` is valueless.
- *Remarks*: This constructor does not require that `is_move_constructible_v<T>` is true.

```
polymorphic(
    std::allocator_arg_t, const Allocator& alloc, polymorphic&& other) noexcept;
```

- *Preconditions*: `other` is not valueless and `Allocator` meets the *Cpp17Allocator* requirements.
- *Effects*: Equivalent to the preceding constructor except that the allocator is initialized with `alloc`.
- *Postconditions*: `other` is valueless.
- *Remarks*: This constructor does not require that `is_move_constructible_v<T>` is true.

X.Z.4 Destructor [`polymorphic.dtor`]

```
~polymorphic();
```

- *Effects*: If `*this` is not valueless, destroys the owned object.

X.Z.5 Assignment [`polymorphic.assign`]

```
polymorphic& operator=(const polymorphic& other);
```

- *Preconditions*: `other` is not valueless.
- *Effects*: If `*this` is not valueless, destroys the owned object. Then, constructs an owned object using the (possibly derived-type) copy constructor of the object owned by `other`.
- *Postconditions*: `*this` is not valueless.

```
polymorphic& operator=(polymorphic&& other) noexcept(
    allocator_traits<Allocator>::propagate_on_container_move_assignment::value ||
    allocator_traits<Allocator>::is_always_equal::value);
```

- *Preconditions*: `other` is not valueless.
- *Effects*: If `*this` is not valueless, destroys the owned object. Then takes ownership of the object owned by `other`.
- *Postconditions*: `*this` is not valueless. `other` is valueless.

X.Z.6 Observers [polymorphic.observers]

```
constexpr const T& operator*() const noexcept;  
constexpr T& operator*() noexcept;
```

- *Preconditions:* `*this` is not valueless.
- *Effects:* Returns a reference to the owned object.
- *Remarks:* These functions are constexpr functions.

```
constexpr const T* operator->() const noexcept;  
constexpr T* operator->() noexcept;
```

- *Preconditions:* `*this` is not valueless.
- *Effects:* Returns a pointer to the owned object.
- *Remarks:* These functions are constexpr functions.

```
constexpr bool valueless_after_move() const noexcept;
```

- *Returns:* true if `*this` is valueless, otherwise false.

```
constexpr allocator_type get_allocator() const noexcept;
```

- *Returns:* A copy of the Allocator object used to construct the owned object or, if that allocator has been replaced, a copy of the most recent replacement.

X.Z.7 Swap [polymorphic.swap]

```
constexpr void swap(polymorphic& other) noexcept(  
    allocator_traits<Allocator>::propagate_on_container_swap::value ||  
    allocator_traits<Allocator>::is_always_equal::value);
```

- *Preconditions:* `*this` is not valueless, `other` is not valueless.
- *Effects:* Swaps the objects owned by `*this` and `other`.
- *Remarks:* Does not call `swap` on the owned objects directly.

```
constexpr void swap(polymorphic& lhs, polymorphic& rhs) noexcept(  
    allocator_traits<Allocator>::propagate_on_container_swap::value ||  
    allocator_traits<Allocator>::is_always_equal::value);
```

- *Preconditions:* `lhs` is not valueless, `rhs` is not valueless.
- *Effects:* Swaps the objects owned by `lhs` and `rhs`.

X.Z.8 Allocator related traits [polymorphic.traits]

```
template <class T, class Alloc>  
struct std::uses_allocator<polymorphic<T>, Alloc> : true_type {};
```

- *Preconditions:* `Alloc` meets the *Cpp17Allocator* requirements.

X.Z.9 Optional support [polymorphic.optional]

```
template <class T, class Alloc>
class std::optional<polymorphic<T, Alloc>>;
```

The specialization `std::optional<polymorphic<T, Alloc>>` guarantees `size(std::optional<polymorphic<T, Alloc>>) == size(polymorphic<T, Alloc>>)`.

```
// [optional.observe], observers
constexpr const polymorphic<T, Alloc>& operator->() const noexcept;
constexpr polymorphic<T, Alloc>& operator->() noexcept;
```

- *Preconditions:* `*this` contains a value.
- *Returns:* `val`.
- *Remarks:* These functions are `constexpr`. The specialization `std::optional<polymorphic<T, Alloc>>` provides `operator->` that returns a reference to the contained `polymorphic`.

Otherwise, the interface of the specialization is as defined in [optional].

Reference implementation

A C++20 reference implementation of this proposal is available on GitHub at https://www.github.com/jbcoe/value_types.

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A C++20 reference implementation is available on GitHub [https://www.github.com/jbcoe/value_types]

Appendix A: Detailed design decisions

We discuss some of the decisions that were made in the design of `indirect` and `polymorphic`. Where there are multiple options, we discuss the advantages and disadvantages of each.

Two class templates, not one

It is conceivable that a single class template could be used as a vocabulary type for an indirect value-type supporting polymorphism. However, implementing this would impose efficiency costs on the copy constructor when the owned object is the same type as the template type. When the owned object is a derived type, the copy constructor uses type erasure to perform dynamic dispatch and call the derived type copy constructor. The overhead of indirection and a virtual function call is not tolerable where the owned object type and template type match.

One potential solution would be to use a `std::variant` to store the owned type or the control block used to manage the owned type. This would allow the copy constructor to be implemented efficiently when the owned type and template type match. This would increase the object size beyond that of a single pointer as the discriminant must be stored.

For the sake of minimal size and efficiency, we opted to use two class templates.

Copiers, deleters, pointer constructors, and allocator support

The older types `indirect_value` and `polymorphic_value` had constructors that take a pointer, copier, and deleter. The copier and deleter could be used to specify how the object should be copied and deleted. The existence of a pointer constructor introduces undesirable capabilities into the design of `polymorphic_value`, such as allowing the possibility of object slicing on copy when the dynamic and static types of a derived-type pointer do not match.

We decided to remove the copier, delete, and pointer constructor in favour of adding allocator support. A pointer constructor and support for custom copiers and deleters are not core to the design of either class template; both could be added in a later revision of the standard if required.

We have been advised that allocator support must be a part of the initial implementation and cannot be added retrospectively. As `indirect` and `polymorphic` are intended to be used alongside other C++ standard library types, such as `std::map` and `std::vector`, it is important that they have allocator support in contexts where allocators are used.

Pointer-like helper functions

Earlier revisions of `polymorphic_value` had helper functions to get access to the underlying pointer. These were removed under the advice of the Library

Evolution Working Group as they were not core to the design of the class template, nor were they consistent with value-type semantics.

Pointer-like accessors like `dynamic_pointer_cast` and `static_pointer_cast`, which are provided for `std::shared_ptr`, could be added in a later revision of the standard if required.

Comparisons and hashing

We support comparisons and hashing for `indirect` but not `polymorphic`. This is because comparing and hashing polymorphic types is not a uniquely solved problem, though it could well be implemented by adding suitable member functions to the base class. Rather than impose the signatures of these member functions upon users of `polymorphic`, we decided to leave hashing and comparison unsupported but implementable by users.

For `indirect`, in the case where the owned object `T` is hashable or comparable, `indirect<T>` is hashable or comparable by forwarding the hash or comparison to the owned object.

Implicit conversions

We decided that there should be no implicit conversion of a value `T` to an `indirect<T>` or `polymorphic<T>`. An implicit conversion would require using the free store and memory allocation, which is best made explicit by the user.

```
Rectangle r(w, h);
polymorphic<Shape> s = r; // error
```

To transform a value into `indirect` or `polymorphic`, the user must use the appropriate constructor.

```
Rectangle r(w, h);
polymorphic<Shape> s(std::in_place_type<Rectangle>, r);
assert(dynamic_cast<Rectangle*>(&s) != nullptr);
```

Explicit conversions

The older class template `polymorphic_value` had explicit conversions, allowing construction of a `polymorphic_value<T>` from a `polymorphic_value<U>` where `T` was a base class of `U`.

```
polymorphic_value<Quadrilateral> q(std::in_place_type<Rectangle>, w, h);
polymorphic_value<Shape> s = q;
assert(dynamic_cast<Rectangle*>(&s) != nullptr);
```

Similar code cannot be written with `polymorphic` as it does not allow conversions between derived types:

```
polymorphic<Quadrilateral> q(std::in_place_type<Rectangle>, w, h);
polymorphic<Shape> s = q; // error
```

This is a deliberate design decision. `polymorphic` is intended to be used for ownership of member data in composite classes where compiler-generated special member functions will be used.

There is no motivating use case for explicit conversion between derived types outside of tests.

A converting constructor could be added in a future version of the C++ standard.

Small object optimization for `polymorphic`

`polymorphic` could be designed to include a small object optimization like that used in `std::string`. A small object optimization uses a buffer to potentially store the owned object and avoid allocating memory. This would make move construction more complicated as the owned object must be moved from one buffer to another, potentially invoking allocations if the owned object's move constructor allocates memory.

As designed, `polymorphic<T>` does not require that `T` (or constructed classes of type `U` derived from `T`) are move constructible or move assignable for `polymorphic<T>` to be move constructible or move assignable.

A `polymorphic` value type with a small buffer optimization that did not allocate a control block for the owned object would need to be a different type. It is not possible to add a small object optimization to `polymorphic` without making breaking changes. There may be a case for the addition of `small_polymorphic<T, N>` similar to `llvm::SmallVector<T, N>`, but we are not proposing its addition here.